

① Square

[Algorithm [Square pattern]]

- 1) Start
- 2) Input the value of 'n'
- 3) Initialize outer loop $i = 1$ to n [for rows]
- 4) Initialize inner loop $j = 1$ to n [for columns]
- 5) Print +
- 6) Repeat step 4 until all columns are printed
- 7) Print new line
- 8) Repeat steps 3-7 until all rows are printed
- 9) Stop.

C++

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    int i, j;
```

```
    for (i = 1; i <= 5; i++) {
```

```
        for (j = 1; j <= 5; j++) {
```

```
            cout << "+";
```

```
        }
```

```
        cout << endl;
```

```
    }
```

```
    return 0;
```

```
}
```

C

```
#include <stdio.h>
```

```
int main () {
```

```
    int i, j;
```

```
    for (i = 1; i <= 5; i++) {
```

```

for (j=1; j<=5; j++) {
    print (" * ");
}
printf("\n");
}
return 0;
}

```

Output

```

* * * * *
* * * * *
* * * * *
* * * * *
* * * * *

```

2) Hollow square

```

#include <iostream>
using namespace std;

```

```

int main() {
    int i, j;
    for (i=1; i<=5; i++) {
        for (j=1; j<=5; j++) {
            if (i==1 || i==5 || j==1 || j==5)
                cout << " * ";
            else
                cout << " ";
        }
        cout << endl;
    }
    return 0;
}

```

C

```

#include <stdio.h>

```

```

int main() {
    int i, j;

```

```

    for (i=1; i<=5; i++) {
        for (j=1 || i==5 || j==1 || j==5)

```

```

printf("%s")
else
printf("|" " ");
}
printf("\n");
}
return 0;
}

```

```

* * * * *
*       *
*       *
*       *
* * * * *

```

③ Triangle

C++

```

*
* *
* * *
* * * *
* * * * *

```

```

#include <iostream>
using namespace std;

```

```

int main() {
    int i, j;

    for (i = 1; i <= 5; i++) {
        for (j = 1; j <= i; j++) {
            cout << " ";
        }
        cout << endl;
    }
    return 0;
}

```

C

```

#include <stdio.h>

```

```

int main() {
    int i, j;

    for (i = 1; i <= 5; i++) {
        for (j = 1; j <= i; j++) {
            printf("%s");
        }
        printf("\n");
    }
    return 0;
}

```

4) Inverted Triangle

C++

```

* * * * *
* * * *
* * *
* *
*

```

```

#include <iostream>
using namespace std;
int main() {
    int i, j;

```

C language

```

#include <stdio.h>

```

```

int main() {
    int i, j;

```

```

    for (i = 5; i >= 1; i--) {
        for (j = 1; j <= i; j++) {
            printf(" ");
        }
        printf("\n");
    }
    return 0;
}

```

```

    for (i = 5; i >= 1; i--) {
        for (j = 1; j <= i; j++) {
            cout << " ";
        }
        cout << endl;
    }
    return 0;
}

```

5) Pyramid

C++

```

#include <iostream>

```

```

using namespace std;

```

```

int main() {
    int i, j, k;

```

```

    for (i = 1; i <= 5; i++) {
        for (j = 1; j <= 5 - i; j++) {
            cout << " ";
        }

```


for (k=1; k<=2*i-1; k++)
cout << "x" ;

cout << endl;

}
return 0;
}

#include <stdio.h>

③

int main() {

int i, j, k;

for (i=1; i<=5; i++) {

for (j=1; j<=5-i; j++)

printf(" ");

for (k=1; k<=2*i-1; k++)

printf("x");

printf("\n");

} return 0;

dry run

```

      x
     x x
    x x x
   x x x x
  x x x x x
 
```

6) Inverted Pyramid

#include <iostream>

using namespace std;

int main() {

int i, j, k;

for (i=5; i>=1; i--) {

for (j=1; j<=5-i; j++)

cout << " " ;

for (k=1; k<=2*i-1; k++)

cout << "x";

cout << endl;

} return 0;

```

x x x x x
 x x x x
  x x x
   x x
    x
 
```

C (Language)

```

#include <stdio.h>
int main () {
    int i, j, k;

    for (i = 5; i >= 1; i--) {
        for (j = 1; j <= 5 - i; j++)
            printf(" ");
        for (k = 1; k <= 2 * i - 1; k++)
            printf("x");
        printf("\n");
    }
    return 0;
}

```

7) Half Diamond

```

#include <iostream>
using namespace std;

```

```

int main () {
    int i, j;

    for (i = 1; i <= 5; i++) {
        for (j = 1; j <= i; j++)
            cout << "x";
        cout << endl;
    }

    for (i = 4; i >= 1; i--) {
        for (j = 1; j <= i; j++)
            cout << "x";
        cout << endl;
    }

    return 0;
}

```

C Language

#include <stdio.h>

```
int main() {  
    int i, j, k;
```

```
    for (i=1; i<=5; i++) {
```

```
        for (j=1; j<=5-i; j++)  
            printf(" ");
```

```
        for (k=1; k<=2*i-1; k++)  
            printf("\n");
```

```
    }
```

```
    for (i=4; i>=1; i--) {
```

```
        for (j=1; j<=5-i; j++)  
            printf(" ");
```

```
        for (k=1; k<=2*i-1; k++)  
            printf("x");
```

```
        printf("\n");
```

```
    }
```

```
    return 0;
```

```
}
```

⑧ Diamond

Cpp

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    int i, j, k;
```

```
    // Upper part
```

```
    for (i = 1; i <= 5; i++) {
```

```
        for (j = 1; j <= 5 - i; j++)
```

```
            cout << " ";
```

```
        for (k = 1; k <= 2 * (i - 1); k++)
```

```
            cout << "x";
```

```
        cout << endl;
```

```
    }
```

```
    // Lower part
```

```
    for (i = 4; i >= 1; i--) {
```

```
        for (j = 1; j <= 5 - i; j++)
```

```
            cout << " ";
```

```
        for (k = 1; k <= 2 * (i - 1); k++)
```

```
            cout << "x";
```

```
        cout << endl;
```

```
    }
```

```
    return 0;
```

```
}
```


diamond

C

```
#include <stdio.h>
```

```
int main () {
```

```
    int i, j, k;
```

```
    for (i=1; i <= 5; i++) {
```

```
        for (j=1; j <= 5-i; j++)
```

```
            printf(" ");
```

```
        for (k=1; k <= 2*i-1; k++)
```

```
            printf("x");
```

```
        printf("\n");
```

```
    }
```

```
    for (i=4; i >= 1; i--) {
```

```
        for (j=1; j <= 5-i; j++)
```

```
            printf(" ");
```

```
        for (k=1; k <= 2*i-1; k++)
```

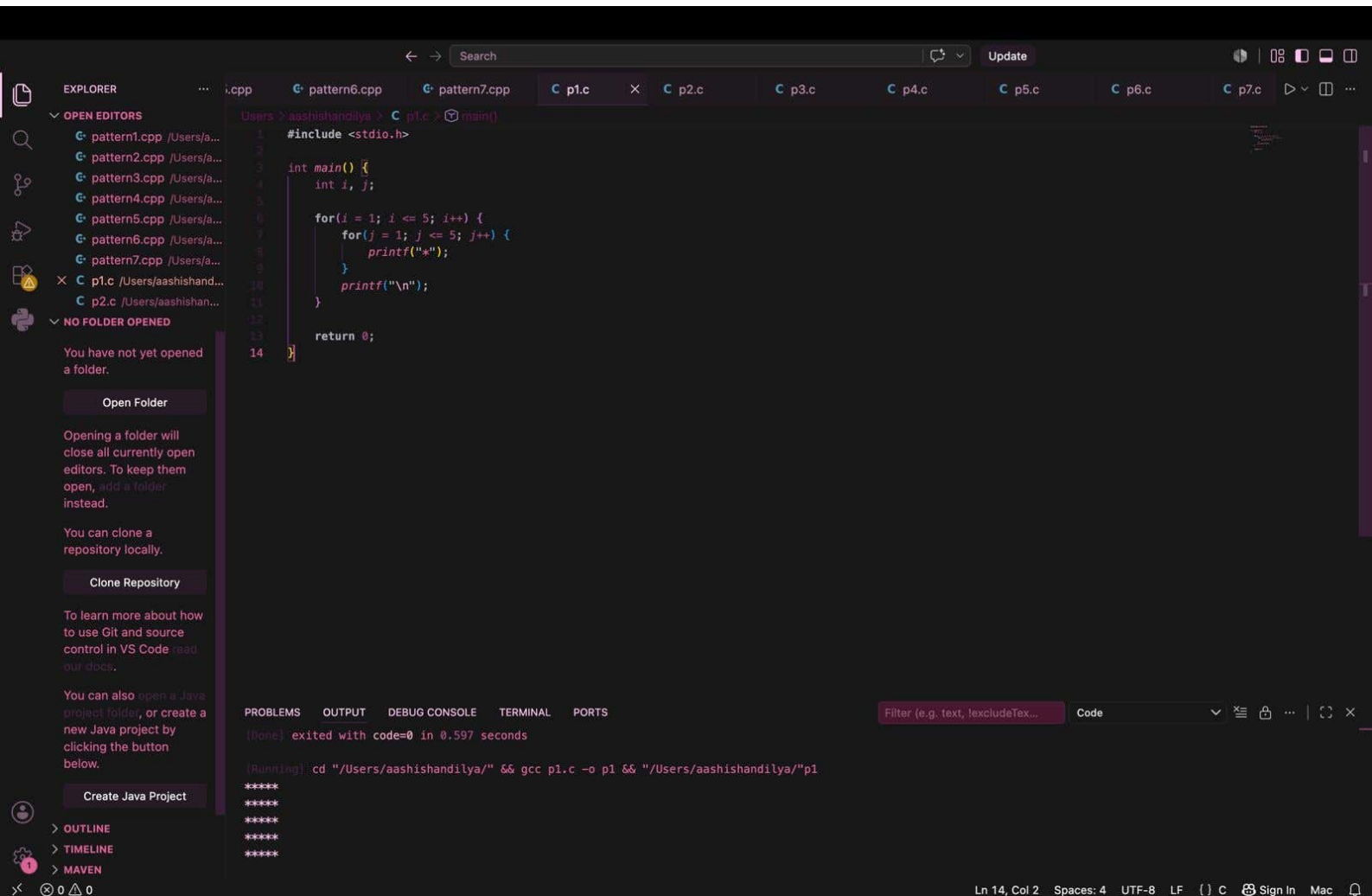
```
            printf("x");
```

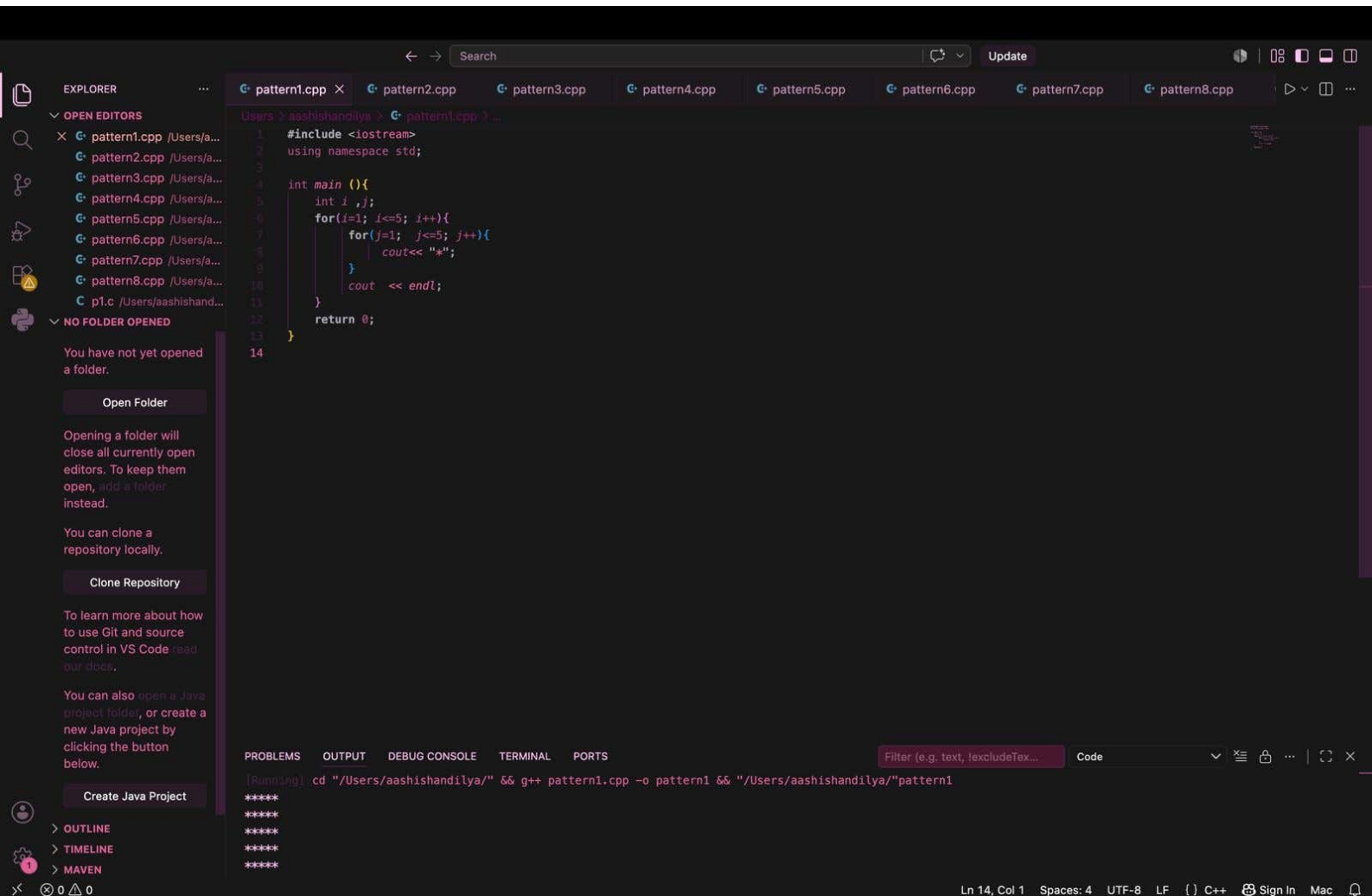
```
        printf("\n");
```

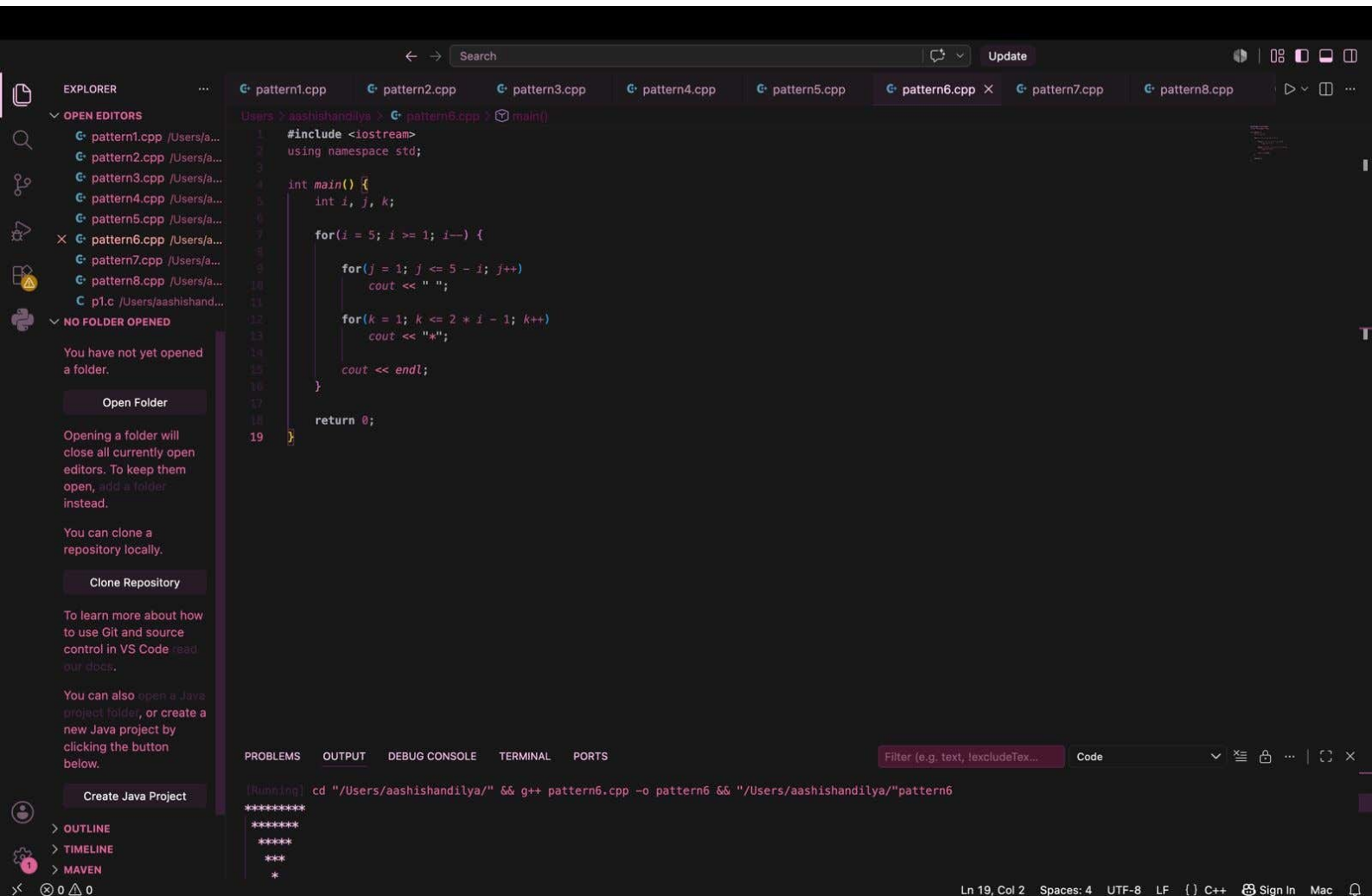
```
    }
```

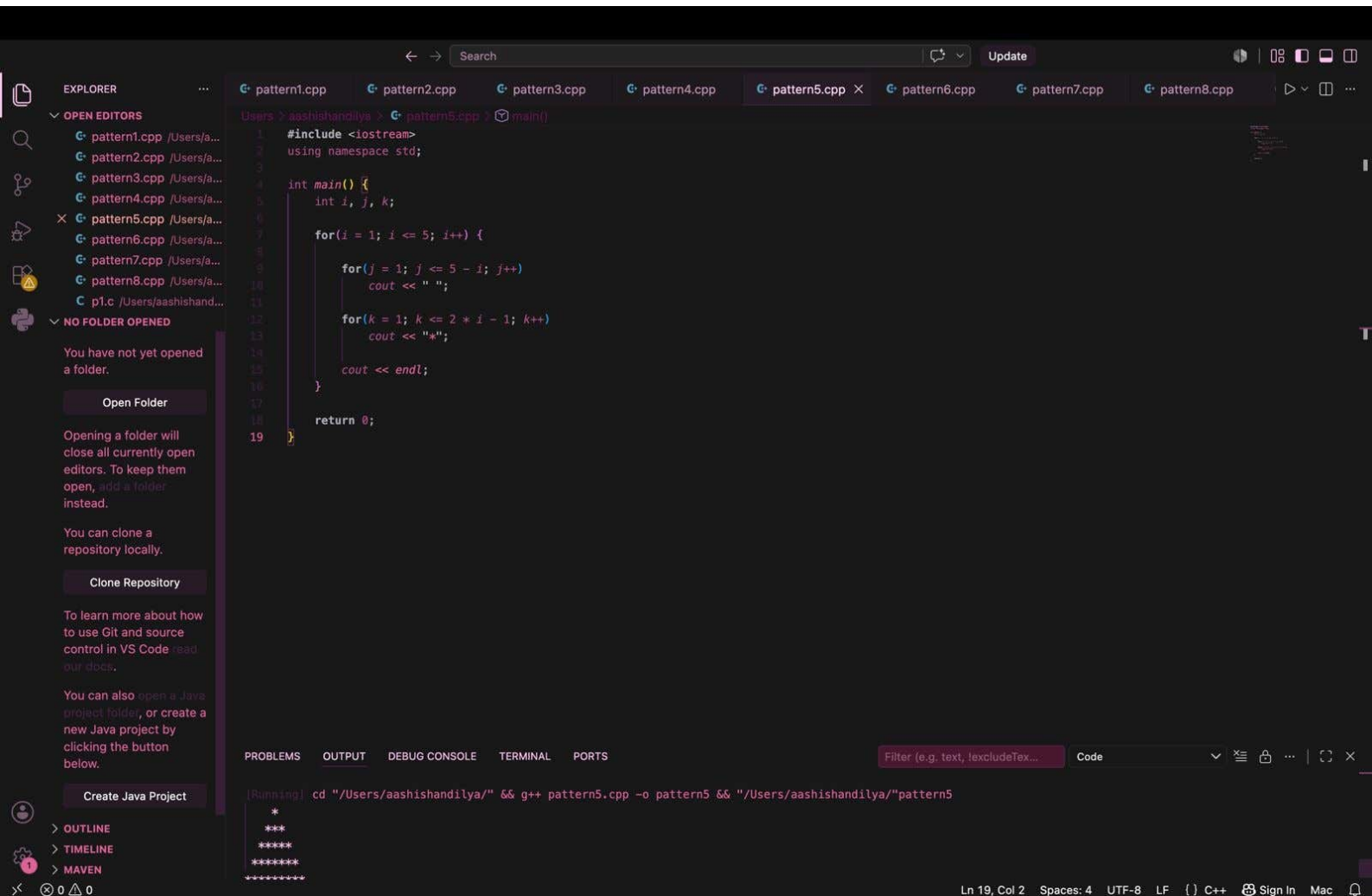
```
    return 0;
```

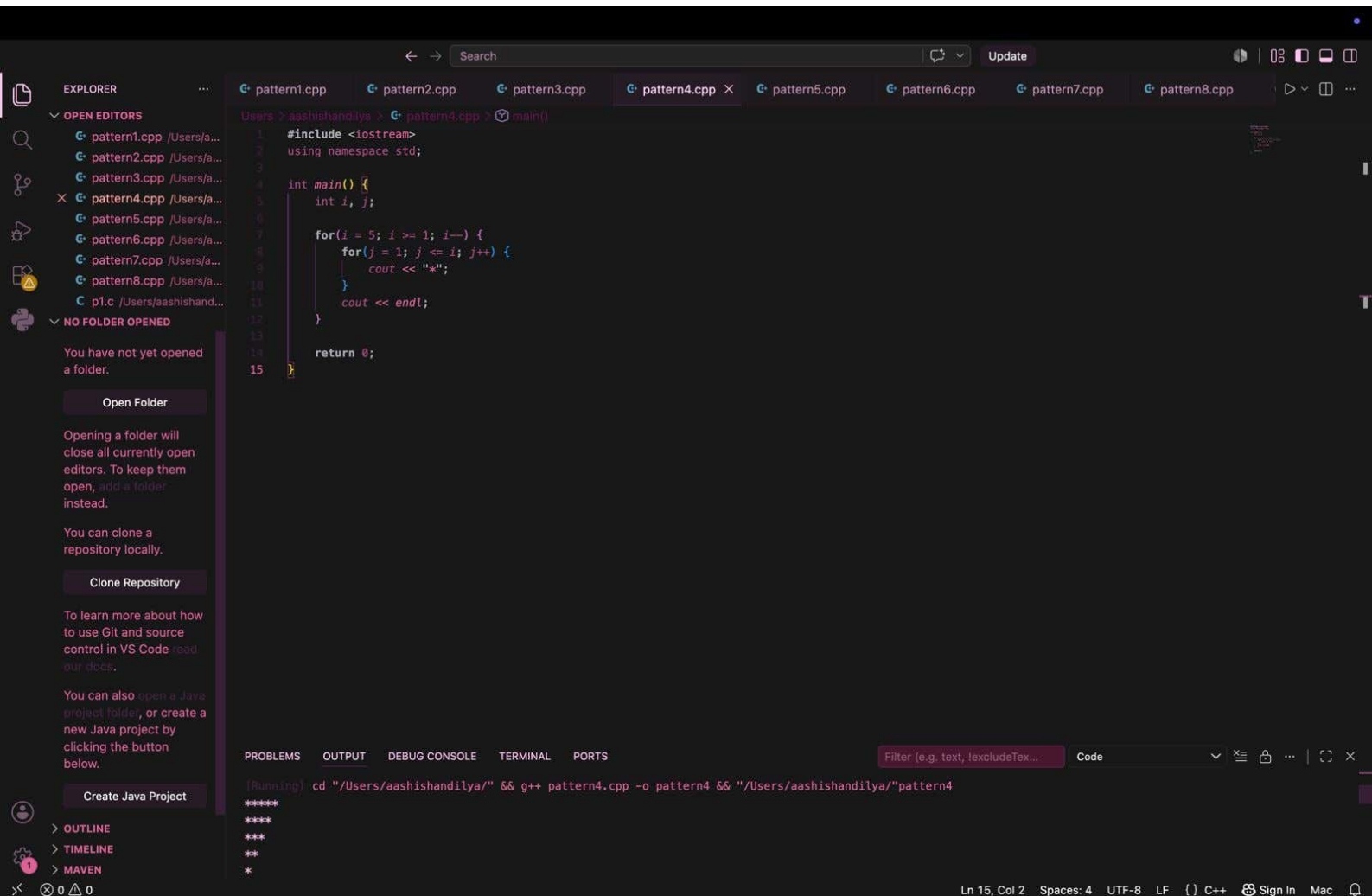
```
}
```

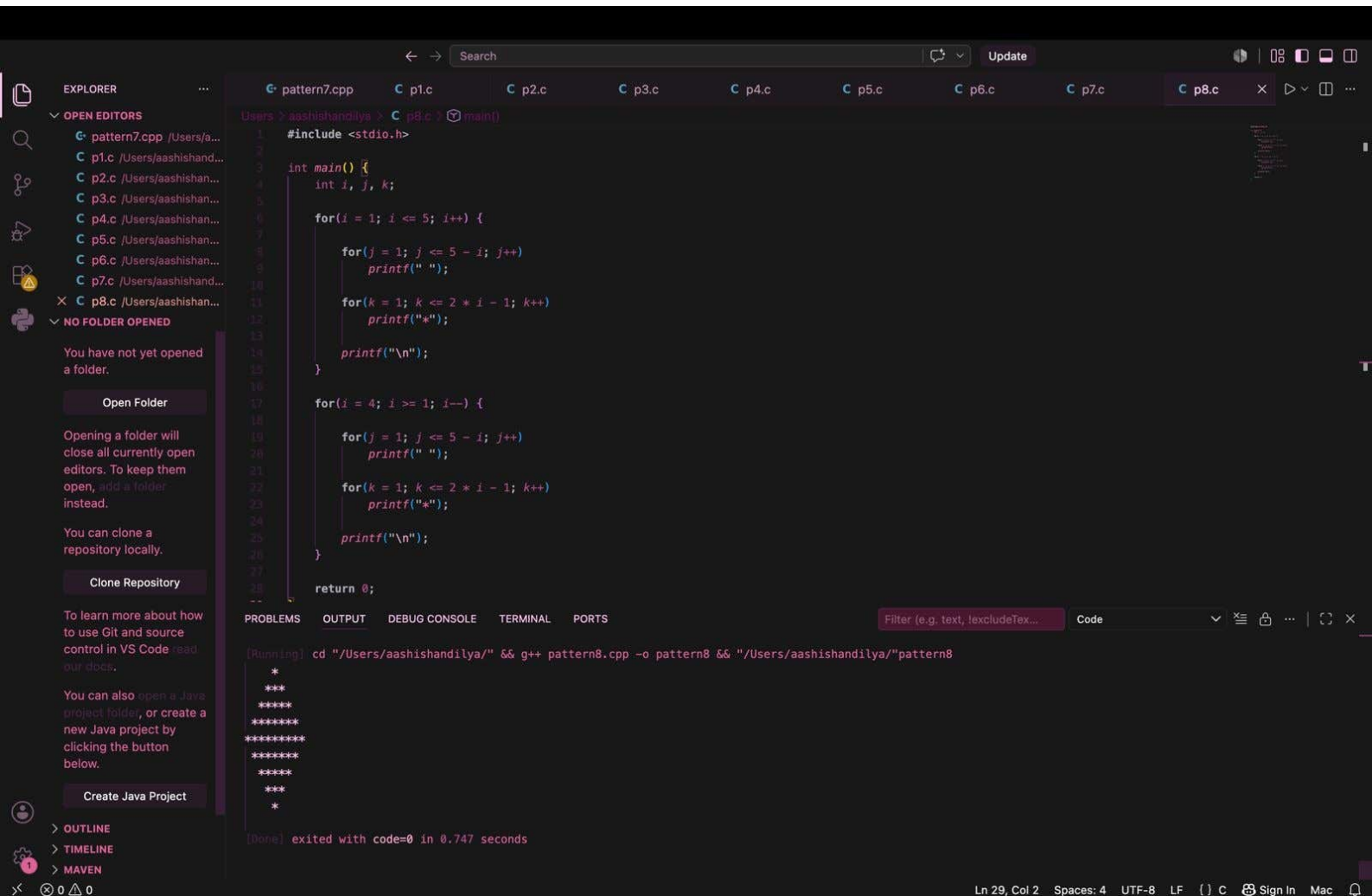


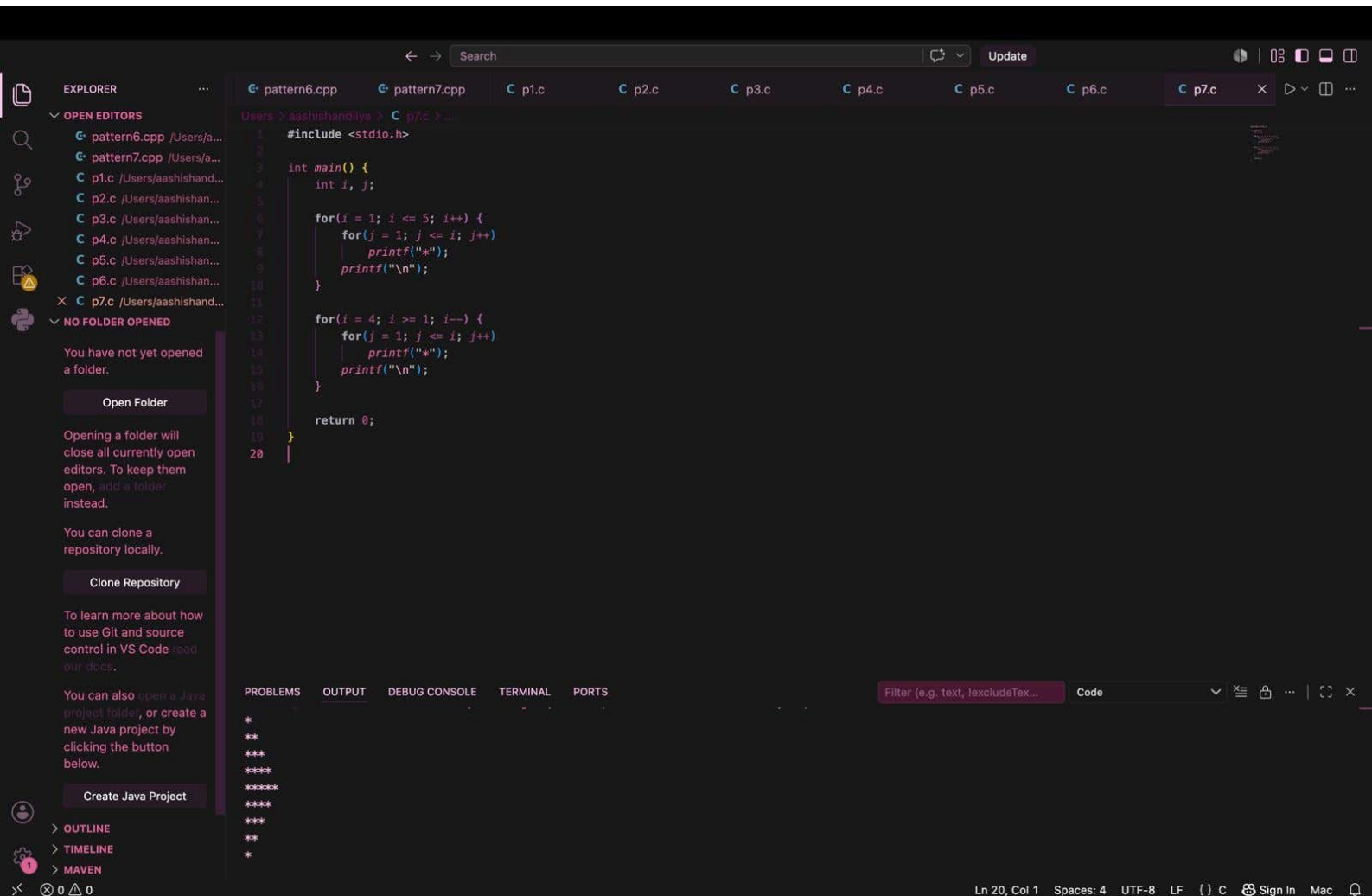


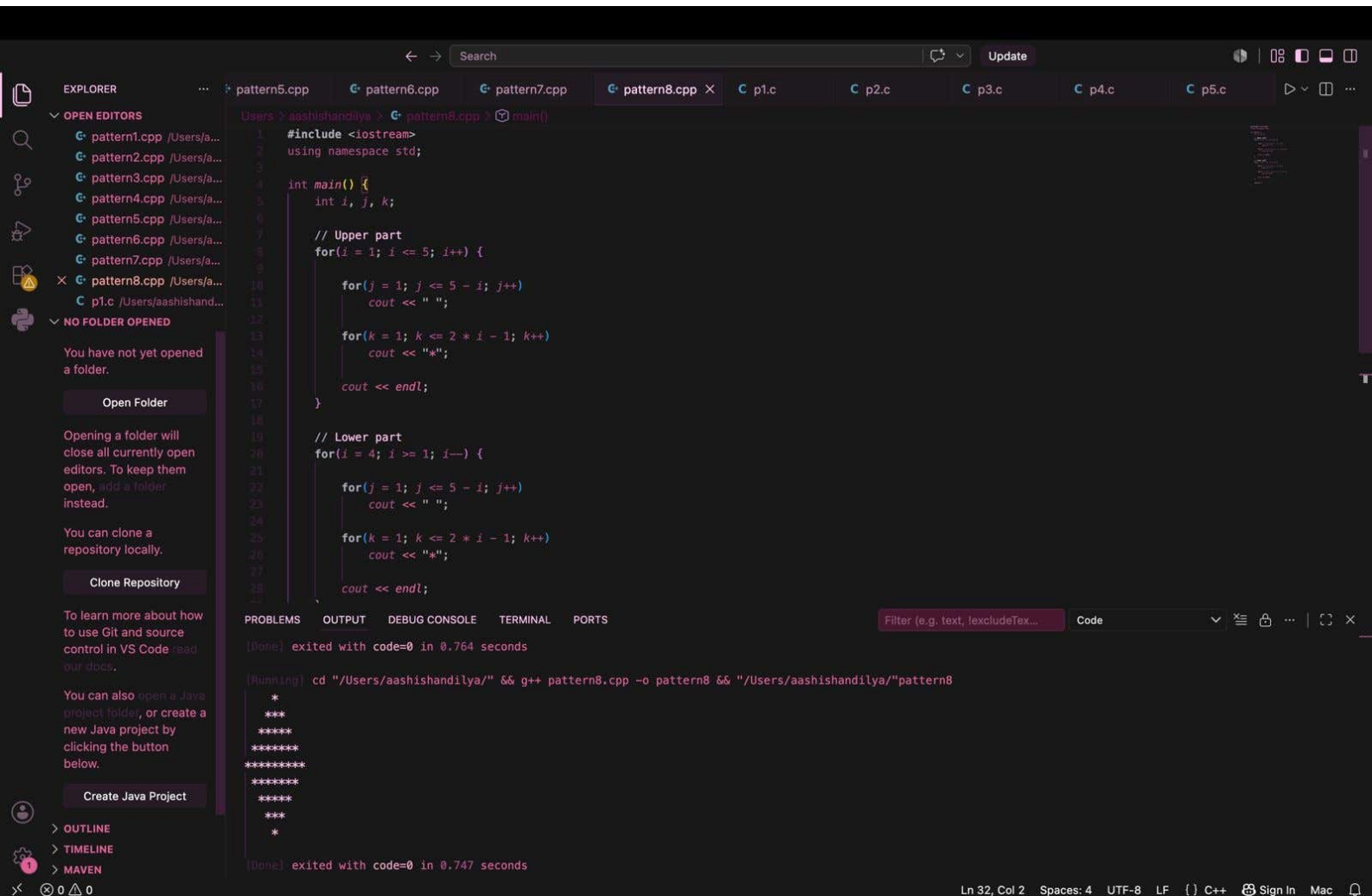


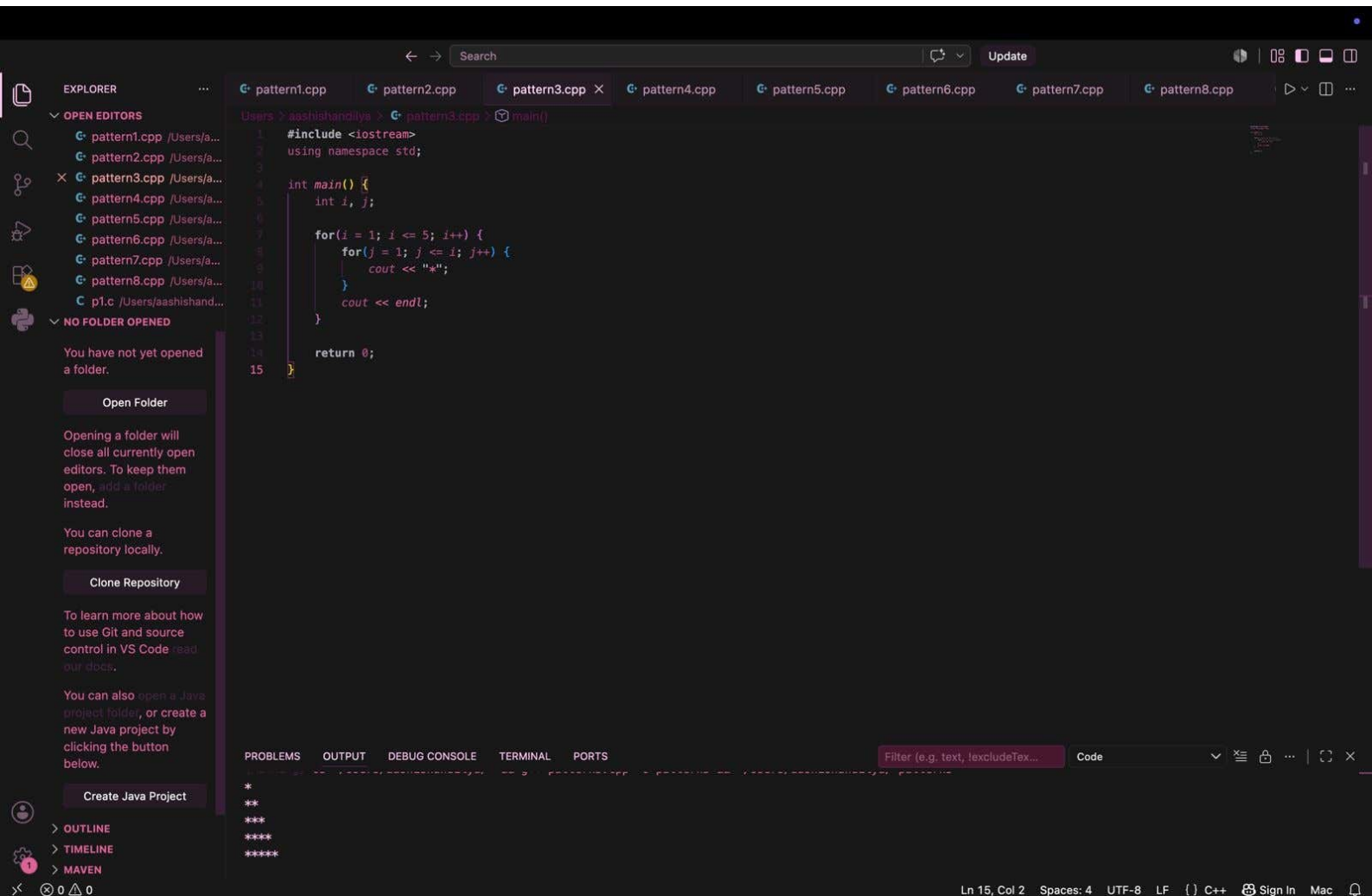


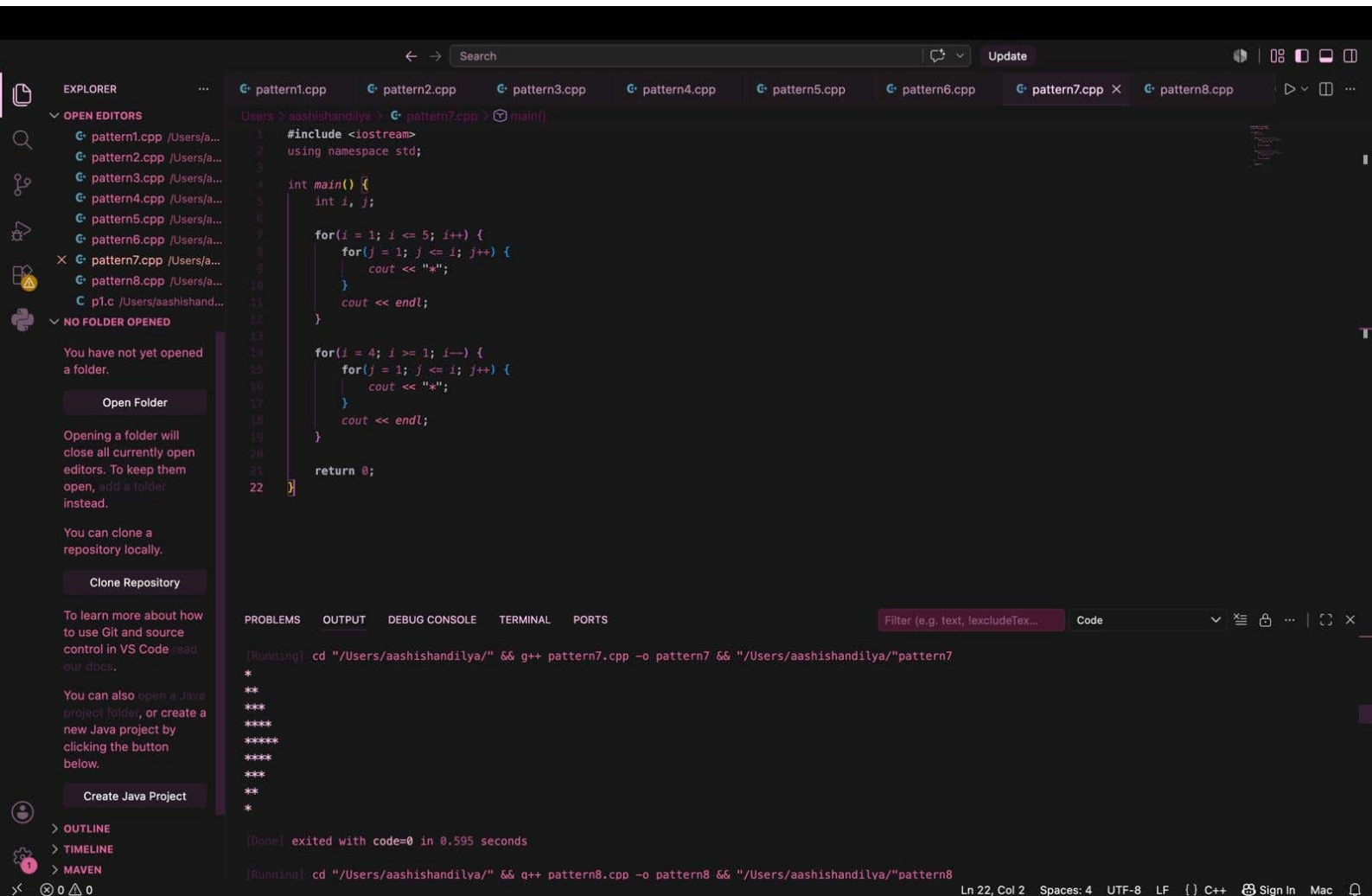


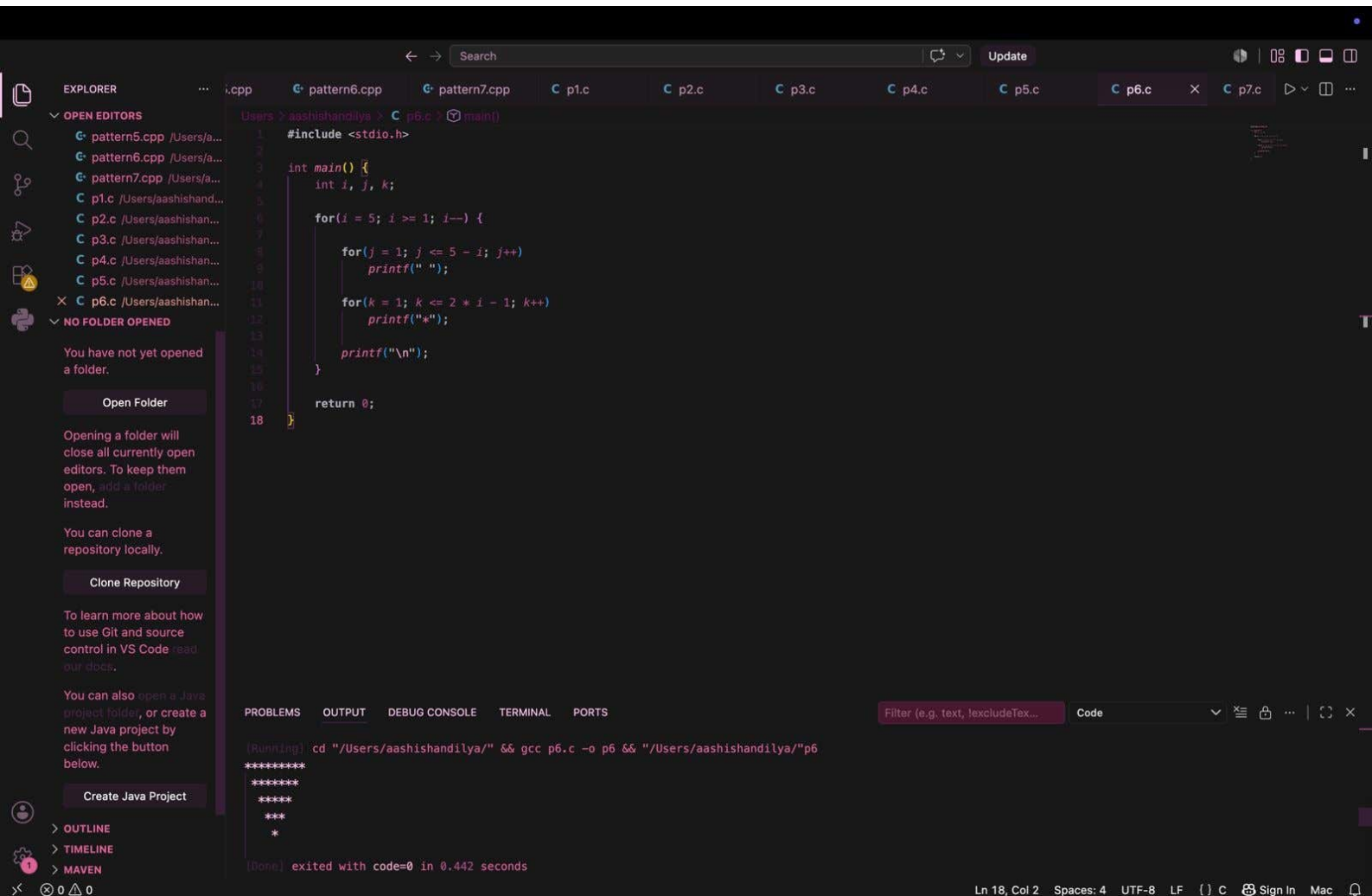












EXPLORER

OPEN EDITORS

pattern1.cpp /Users/a...

pattern2.cpp /Users/a...

pattern3.cpp /Users/a...

pattern4.cpp /Users/a...

pattern5.cpp /Users/a...

pattern6.cpp /Users/a...

pattern7.cpp /Users/a...

pattern8.cpp /Users/a...

p1.c /Users/aashishand...

NO FOLDER OPENED

You have not yet opened a folder.

Open Folder

Opening a folder will close all currently open editors. To keep them open, add a folder instead.

You can clone a repository locally.

Clone Repository

To learn more about how to use Git and source control in VS Code read our docs.

You can also open a Java project folder, or create a new Java project by clicking the button below.

Create Java Project

OUTLINE

TIMELINE

MAVEN

pattern1.cpp

pattern2.cpp

pattern3.cpp

pattern4.cpp

pattern5.cpp

pattern6.cpp

pattern7.cpp

pattern8.cpp

Users > aashishandilya > pattern2.cpp > main()

```
1  #include <iostream>
2  using namespace std;
3
4  int main () {
5      int i, j;
6      for(i=1; i<=5; i++){
7          for(j=1; j<=5; j++){
8
9              if(i==1 || i==5 || j==1 || j==5)
10                 cout << "*";
11             else
12                 cout << " ";
13
14         }
15         cout << endl;
16     }
17     return 0;
18 }
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

Filter (e.g. text, excludeTex...

Code

[Running] cd "/Users/aashishandilya/" && g++ pattern2.cpp -o pattern2 && "/Users/aashishandilya/"pattern2

* *

* *

* *

Ln 17, Col 17

Spaces: 4

UTF-8

LF

{ } C++

Sign In

Mac

